frustrated by the inability to find "things signified" behind the notions of negative and imaginary numbers, which therefore looked absurd to him. By 1828 he became "greatly dissatisfied with the phrases, if not the reasonings, of even very eminent analysts". He believed that in order to "clear away the metaphysical stumbling-blocks that beset the entrance of analysis", one needed either to discard negative and imaginary numbers or to explain their "true sense" [**17**, 2:143, 1:304]. Hamilton soon learned of the so-called Argand diagram representing the complex number as a point on a plane, with its real and imaginary parts plotted on two rectangular axes. Inspired by this geometrical representation, he began looking for an algebraic representation of complex numbers for which all valid operations could be defined and soon developed the concept of algebraic couples. Echoing the Kantian vision of algebra as a science of time, Hamilton viewed these number couples as "steps" in time, rather than magnitudes, and he interpreted negative signs as reversals of temporality [**43**, p. 281]. He sought a new foundation of algebra in the intuitive notion of pure or mathematical time: the moment was to algebra what the point was to geometry, time intervals were finite straight lines, and an indefinite straight line represented Time itself. Being awfully busy prevented Hamilton from pursuing this promising line of reasoning further. As he remarked, "Time is needed, with all its gross reality of hours and days, even to write upon Pure Time" [**17**, 2:144].

The 2-dimensional representation of complex numbers also inspired Hamilton to seek hypercomplex numbers related to the "real" 3-dimensional space, or "triplets", as he called them. "Triplets" fit nicely with his philosophical interest in the idealistic triad "thesis-antithesis-synthesis" [**20**]. Adding triplets was easy, but they stubbornly refused to multiply. They did it for a good reason, for, as Frobenius would prove only after Hamilton's death, no such algebra was possible. In desperation, Hamilton tried ordered sets of four numbers, or "quaternions". They agreed to multiply only if the commutative law was lifted. This realization came to Hamilton on October 16th, 1843, as he was walking with his wife along the Royal Canal in Dublin. "I then and there felt the galvanic circuit of thought *close*; and the sparks which fell from it were the *fundamental equations*"